1) g is a parent of y where y is a parent and x is a parent of y.

$\forall y \exists y \exists x(y \neq x \wedge \forall$ ~~$\forall g \exists y(y_1 \neq y_2 \wedge \forall x(P(x,y) \to x = y))$~~

fore every grandparent there exists a parent where no two are the same and every parent of a parent is a grandparent

$$\forall_g \exists_y \exists_x (y_1 \neq y_2 \wedge \forall_x (P(x,y) \to x = y) \wedge \forall_g (G(g,y) \to g = y)$$

2. A and B are sets prove. if A is a subset of B then A intersects B = A

proof. $x \in A$ let this be arbitrary. by definition since x is in the set of A and A is the subset of B then $x \in B$. since both $x \in A$ and $x \in B$ are true, then x is in the intersection of A and B.

3. Base case: $n=1$ $\frac{1}{1(4)} = \frac{1}{4}$ ✓

Inductive case: for every positive integer $k \in \mathbb{N}_{n \geq 0}$ is true then $P(k+1)$ is true. let $n = k$ so $\frac{k}{3k+1}$

let $n = k+1$ so $\frac{k+1}{3k+4}$ if $k=1$ then $\to \frac{1}{4} + \frac{1}{3(1)+4}\left(\frac{1}{3(1)+1}\right) = \frac{1}{4} + \frac{1}{4 \cdot 7}$

Thus, the statement holds for $k+1$.

4. $a_n = 4a_{n-1} - 4a_{n-2} + 3^n$ for $n \geq 2$ $a_0 = 6$ $a_1 = 8$

$$\sum a_n x^{n+2} = 4\left(\sum a_{n-1} x^{n+2}\right) - 4\left(\sum a_{n-2} x^{n+2}\right) + \sum 3^n x^{n+2}$$

$$x^2 G(x) = 4x\,G(x) - 4\,G(x) + 3x^2$$

$$G(x) = \left(\frac{3x^2}{\quad}\right) \Big/ (x-2)^2 = \frac{3x^2}{(x-2)^2}$$

(Ran out of time here) just going to guess

$A = -1$ $B = 1$

$G(x) = \frac{-1}{x-2} + \frac{1}{x-2}$

5. Any for Alice   n identical balloons to 3 distinct children
atmost 2 for Ben
Chloe even #

For Alice $- \; 1+x+x^2+\ldots = \frac{1}{1-x}$

For Ben $- \; 1+x^1+x^2 = \frac{1-x^3}{1-x}$

For Chloe $- \; 1+x^2+x^4+\ldots = \frac{1}{1-x^2}$

$$\frac{1}{1-x} + \frac{1-x^3}{1-x} + \frac{1}{1-x^2} \Rightarrow \sum x^n + \sum (1-x^3) x^n + \sum x^n$$

$$\downarrow$$

$$\sum 2x^n + \sum (1-x^3) x^n$$

so there are $2(1-x^n)$ ways to distribute the balloons (?)

6. $A(n) = 2A(n/2) + 1$

$a=2 \quad b=2$ After work we get $A(n) = \frac{3}{2}n + 1$ (ran out of time) sorry im guessing here

$a < b^d = O(n^d)$

the runtime for this algorithm is $A(n) = O(n)$